

Forgery Attack on a Secure Data Sharing for Industrial IoT

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Abstract—In recent years and with the emergence of the industrial revolution, the secure data sharing schemes have been developed in IoT platforms and have been recognized as a hot topic in industry and academia. These schemes enable IoT devices to securely share their sensed data in industrial environments with clients through an appropriate infrastructure and intermediary entities. The research conducted in this field shows the existence of various security challenges and solutions. Data privacy, data authentication, fairness and accountability are some of the most important security features presented. Recently, in paper [Sengupta-Ruj-Bit, TNSM 2023] proposed a secure sharing scheme and claimed that it covers all the mentioned security features even when entities collude with each other. In this paper, we investigate the security analysis of this scheme, and show that it does not cover the claimed fairness property. Therefore, the mentioned scheme is vulnerable and cannot be used as a valid scheme in real-world applications.

Index Terms—Fog computing, Internet of things, Fairness, Data privacy, Data authenticity



1 INTRODUCTION

The fourth industrial revolution, also known as Industry 4.0, refers to the development and automation of manufacturing and industrial processes through technologies such as: internet of things (IoT), cloud computing, big data, artificial intelligence (AI) and machine learning. Some of the most important achievements of this revolution are: increased efficiency and productivity, improved quality, enhanced safety, and reduced costs [1–3].

The secure data sharing in Industrial Internet of Things (IIoT) is one of the hot topics raised in the Industry 4.0. The reason is that the data generated by IoT devices provides significant potential for the goals pursued by the industrial revolution [4–6]. The scenario model of such a scheme is as follows: at first, IoT devices generate and collect data from industrial environments, and then, through an intermediary entity, called a fog, the data is processed and aggregated, and finally outsourced to the storage space of a cloud server.

Many papers have been written in this area [7–15], however, because our main goal is security analysis of one of these papers, only those that are closer to this paper are described below.

Manzoor et al. [11], using a combination of blockchain and cloud server, have proposed a secure data sharing scheme for IoT platforms. Their scenario model considers four entities: IoT devices, client, blockchain, and cloud server, in which IoT devices share their data with clients via storage space provided by a cloud server. All transactions and interactions between these entities are done

through smart contracts employed on the blockchain. Their scheme guarantees security features such as: confidentiality, anonymity, auditability, and fairness using cryptographic tools such as: public-key encryption schemes, cryptographic hash functions, re-encryption schemes, and computationally hardness problems such as: the discrete logarithm and the computational Diffie-Hellman.

Gao et al. [12] have proposed another scheme for secure data sharing of IoT devices, which includes four entities: client, IoT devices, SDN network, and cloud server. In such a setting, the process of generating and collecting data is done by IoT devices, and also outsourcing and sharing it on a cloud server for clients is done by SDN network. Their scheme guarantees security features such as: data confidentiality and data integrity, as well as authentication of IoT devices using cryptographic tools such as: bilinear map groups, cryptographic hash functions, and re-encryption schemes.

Singh et al. [13] have presented a general framework for secure data sharing in the IoT platform. Within this framework, entities such as: cloud servers, IoT devices, security gateways, and blockchain are defined. The security gateways are the interface between the cloud server and IoT devices, and their main task is to provide security and detect malicious behavior of entities. Transactions and interactions are also audited automatically by the blockchain and smart contracts deployed on it. In summary, with the facilities used, this framework provides security features such as confidentiality, authentication, access control, and auditing.

Dziembowski et al. [14] have proposed an efficient protocol for the secure exchange of digital goods using smart contracts, and have introduced the secure exchange of data generated by IoT devices as one of their applications. Their protocol includes the following entities: a cloud server for storing and sharing data, a blockchain with multiple smart contracts for auditing, along with two entities: the data producer and the client. The security features supported

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TABLE 1: The comparison of secure data sharing schemes for Iot platform

Scheme	Contributed Entities	Crypto. Tools	Security Features							
			Data privacy	Data authenticity	Fairness	Accountability	Anonymity	Authentication	Authorization	Collusion-resistant
Manzoor et al. [11]	IoT device, Client Blockchain, Cloud	Public-key encryption, Hash function Re-encryption scheme, DLP problem CDH problem	✓	×	✓	✓	✓	×	×	×
Gao et al. [12]	IoT device, Client Cloud, SDN network	Bilinear group, Hash function Re-encryption scheme	✓	✓	×	×	×	✓	×	×
Singh et al. [13]	IoT device, Blockchain Security gateway, Cloud	Nothing (this is an abstract framework).	✓	×	×	✓	×	✓	✓	×
Dziembowski et al. [14]	IoT device, Client Blockchain, Cloud	Hash function, Commitment scheme Symmetric-key encryption, Merkle tree	✓	×	✓	×	×	×	×	×
Sengupta et al. [15]	TA, IoT device, Cloud Blockchain, Fog, Client	Bilinear group, Symmetric-key encryption Digital signature, Public-key encryption	✓	✓	☐	✓	×	×	×	✓

Note: A boxed item indicates that the claimed security feature has been violated.

by this protocol include data confidentiality and fairness, which are provided using cryptographic tools such as hash functions, commitment schemes, symmetric-key encryption schemes, and merkle trees.

Recently, Sengupta et al. [15] have introduced a secure data sharing scheme for data generated by IoT devices in industrial environments. Their scheme consists of six entities: Trusted authority, IoT devices, Fog, Cloud server, Client and Blockchain. The authors claim that, while allowing collusion between entities that was neglected in previous schemes, their scheme covers security features such as: data authenticity, data privacy, accountability and fairness. To support the mentioned features, this scheme uses smart contracts deployed on the blockchain, as well as cryptographic tools such as: bilinear map groups, digital signature schemes, symmetric-key encryption schemes, and public-key encryption schemes.

Table 1 summarizes a detailed overview of the different secure data sharing constructions that we discussed above, in terms of the entities in scenario model, the cryptographic tools used, and the security features.

In this paper, we study the security analysis of the scheme proposed in [15]. In particular, without reducing the level of security and functionality, we rewrite their scheme in such a way that less attention is paid to technical aspects and, conversely, more attention is paid to security aspects and cryptographic tools. It should be noted that this rewriting formalizes the scheme and reduces its complexity for security analysis. Finally, we show that the mentioned scheme does not cover security feature fairness, and therefore is vulnerable and cannot be considered a valid scheme for real-world applications.

1.1 Paper organization

We organize the rest of paper as follows. The scenario model, threat model and security properties related to secure data sharing schemes are explained in Section 2. We provide the notations and cryptographic primitives used throughout the paper in Section 3. Section 4 reviews the FairShare construction proposed in paper [15]. In Section 5, we present a security analysis of the FairShare construction described in Section 4. Finally, Section 6 concludes the paper and points out our future directions.

2 PROBLEM STATEMENT

In this section, we first describe the entities and their interactions with each other in secure data sharing schemes for the IoT platform. Then, we determine the threat model of these schemes by assigning security labels to each entity, and describe the security features required in it.

2.1 System model

Our scenario model includes six entities as follows:

- **Trusted authority (TA):** This entity is responsible for preparing the necessary infrastructure. More precisely, it launches a cloud storage server to outsource data from IoT devices. Also, It launches a blockchain platform and deploys smart contracts on it to monitor and audit interactions and transactions.
- **IoT devices:** They are electronic components with a wide range of sensors, which are used for smart platforms. Their computational and storage resources are limited, and their main task is to collect and send information in small chunks to more powerful entities.
- **Cloud service provider:** An entity that is perceived by other entities to have unlimited storage and computing resources, and its main task is to manage and maintain data outsourced by IoT devices.
- **Fog:** It is an intermediary entity between the cloud server and IoT devices, and its task is to aggregate raw data received from IoT devices and outsource it to the cloud. In terms of computing and storage resources, this entity is far less limited than IoT devices, however, its resources are limited, and to store large amounts of data, it outsources them to the cloud. This entity can be any smart device such as a mobile phone, tablet, or laptop.
- **Client:** In an industrial setting, this entity can be any consumer or distributor who receives services remotely through a public channel.
- **Blockchain platform:** A network of computing nodes and smart contracts that manage and maintain all transactions and interactions between entities by providing an immutable distributed ledger.

The above entities interact with each other in the following four phases:

- **Setup:** In this phase, the only active entity is TA, and it provides the infrastructure needed for other entities.
- **Register:** In this phase, entities Fog, Client, and IoT devices are registered. Upon registration, each of these entities receives its public and private parameters, and needs to share the public parameters.
- **Data outsource:** In this phase, entities Fog, Cloud, and IoT devices interact to outsource data. More precisely, at the beginning, the IoT devices send chunks of data to Fog over several rounds and gradually (within a time window). Next, the fog aggregates the

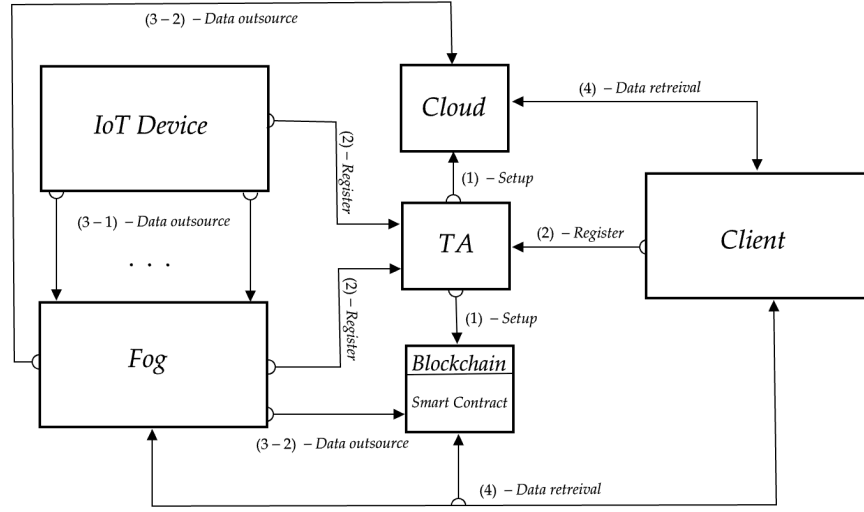


Fig. 1: A view of the discussed system model

received data and stores it in the form of a file on the cloud.

- **Data retrieval and audit:** In this phase, entities Fog, Client, and smart contracts deployed on the blockchain are active. Their interaction is as follows: at first, Fog gradually and in several discrete steps shares metadata with the client and some other entities (appropriate to the need). Throughout the data retrieval phase, smart contracts evaluate and judge the exchange process based on transcripts between entities.

Figure 1 shows an illustration of our system model.

2.2 Threat model

The security label of each entity in our system model is as follows:

- **Trusted authority (TA):** It is known as a fully trusted entity for all entities. It should be noted that this entity is activated and executed only once to launch and register the system.
- **IoT devices:** Since this entity generates the data and has direct access to data, it is considered as a fully trusted entity.
- **Cloud service provider:** It is considered as a malicious entity, meaning that:
 - It tries to extract information beyond what is allowed from encrypted files and transcripts between different entities.
 - It tries to deliver invalid data to clients by changing or generating new files as well as changing transcripts between entities.
- **Fog:** It is considered as a malicious entity, which means that it is trying to get paid by sending an invalid file.
- **Client:** It is considered as a malicious entity, which means that it violates payment despite receiving a valid file.

- **Blockchain platform:** The active nodes in this network (except for a certain threshold of them) are labeled as malicious entities, which means that they try to change and invalidate transactions and interactions between entities.

In such settings, the following three security features are essential for a fair exchange scheme:

- **Data authenticity:** It guarantees that the fog and the client can not forge a valid file in a way to deceive an honest client.
- **Data privacy:** It guarantees that only the authorized parties who have outsourced the file (such as the relevant device and fog), and also the party who purchased the file will be informed of its content, and the other parties will not gain any knowledge about the file.
- **Accountability and fairness:** It guarantees that if the client is able to retrieve the valid file key, the fog receives the specified reward, otherwise it is penalized.

3 PRELIMINARIES

In this section, we review the notations and cryptographic primitives used throughout the paper. It should be noted that readers familiar with these cryptographic primitives can safely ignore Subsection 3.2.

3.1 Notations

Throughout the paper, we denote the security parameter with the symbol λ . Suppose that A denotes an efficient probabilistic algorithm, $y \leftarrow A(x)$ shows that y is the output of algorithm A on x . Similarly, suppose A is an efficient deterministic algorithm, the expression $y := A(x)$ denotes that y is the output of the deterministic algorithm A on x . We use the symbol $\parallel$ to concatenate multiple values into a single value (e.g., $x \parallel y \parallel z = xyz$). We say that a function is negligible, and denote it by negl , if it is smaller than the

inverse of any polynomial in term of the security parameter λ . The symbol $[n]$ is used to represent the set $\{0, 1, \dots, n\}$, where n is a positive integer number.

3.2 Cryptographic primitives

In this subsection, we review several cryptographic primitives that are used in the construction presented in Section 4. Readers familiar with these cryptographic primitives can safely skip this subsection.

Collision-resistant hash function. A function such as:

$$H : \{0, 1\}^* \rightarrow \{0, 1\}^{\text{Poly}(\lambda)}$$

is said to be a collision-resistant hash function if it has the following properties:

- **Efficiency:** For every $x \in \{0, 1\}^*$, the function H is computed in polynomial time in terms of the security parameter λ ,
- **Collision resistant:** It is computationally difficult to find $x \neq y \in \{0, 1\}^*$ such that $H(x) = H(y)$.

Bilinear map. Let $\mathbb{G}_1, \mathbb{G}_T$ are cyclic groups of prime order q , and g_1 is generator for group $\mathbb{G}_1$. A bilinear map is a map such as $e : \mathbb{G}_1 \times \mathbb{G}_1 \rightarrow \mathbb{G}_T$, that satisfies the following conditions:

- **Bilinearity:** $\forall x, y \in \mathbb{Z}_q : e(g_1^x, g_1^y) = e(g_1, g_1)^{xy}$,
- **Non-degeneracy:** $e(g_1, g_1) \neq 1$,
- **Computability:** e can be computed efficiently.

Symmetric-key encryption scheme. It is a triple such as $\text{SKE} = (\text{Gen}, \text{Enc}, \text{Dec})$ that includes key generation, encryption, and decryption algorithms, and has the following properties:

- **CPA security:** For any pair of messages m_0 and m_1 in the message space, the following distributions are computationally indistinguishable:

$$\begin{aligned} & \{k \leftarrow \text{SKE.Gen}(\lambda) : \text{SKE.Enc}_k(m_0)\} \\ & \approx_c \\ & \{k \leftarrow \text{SKE.Gen}(\lambda) : \text{SKE.Enc}_k(m_1)\}. \end{aligned}$$

- **Correctness:** It guarantees that the decryption of any ciphertext corresponding to a given plaintext will lead us to the same plaintext.

Public-key encryption scheme. It is a triple such as $\text{PKE} = (\text{Gen}, \text{Enc}, \text{Dec})$ that includes key generation, encryption, and decryption algorithms, and has the following properties:

- **CPA security:** For any pair of messages m_0 and m_1 in the message space, the following distributions are computationally indistinguishable:

$$\begin{aligned} & \{(pk, sk) \leftarrow \text{PKE.Gen}(\lambda) : \text{PKE.Enc}_{pk}(m_0)\} \\ & \approx_c \\ & \{(pk, sk) \leftarrow \text{PKE.Gen}(\lambda) : \text{PKE.Enc}_{pk}(m_1)\}. \end{aligned}$$

- **Correctness:** It guarantees that the decryption of any ciphertext corresponding to a given plaintext will lead us to the same plaintext.

Digital signature scheme. It is a triple such as $\mathcal{DS} = (\text{Gen}, \text{Sign}, \text{Verify})$ that contains key generation, signature

generation, and verification algorithms, and has the following properties:

- **Unforgeability.** It is not possible to forge a signature for a message without access to the secret keys.
- **Correctness.** It guarantees that any message signed by Sign algorithm will be recognized as a valid signature by Verify algorithm.

4 FAIRSHARE CONSTRUCTION

In this section, we review the secure data sharing scheme FairShare proposed in paper [15], and in section 5 we present a security analysis of this scheme based on the threat model introduced in Subsection 2.2. It should be noted that, without reducing the level of security and functionality, we have rewritten the construction given in [15] and obtain the following construction in such a way that less attention is paid to technical aspects and, conversely, more attention is paid to security aspects.

This construction uses a symmetric-key encryption scheme $\text{SKE} = (\text{Gen}, \text{Enc}, \text{Dec})$ to encrypt data sent from IoT devices to Fog, as well as encrypt data aggregated by Fog. Also, it uses a public-key encryption scheme $\text{PKE} = (\text{Gen}, \text{Enc}, \text{Dec})$ to encrypt the file encryption key, a bilinear group generator GroupGen that efficiently generates a bilinear group description $(\mathbb{G}_1, \mathbb{G}_T, q, g_1, g_T, e)$ by taking a security parameter λ , a digital signature scheme $\mathcal{DS} = (\text{Gen}, \text{Sign}, \text{Verify})$ to maintain the integrity of data exchanged between entities, and a cryptographic hash function $H : \{0, 1\}^* \rightarrow \{0, 1\}^{\text{Poly}(\lambda)}$ to shorten the required signatures. In the following, the description of each of the construction algorithms is explained separately by the phases introduced in Subsection 2.1.

Setup phase. In this phase, there is only one algorithm, called Setup, that it is executed only once at the beginning of the protocol by trusted authority. This algorithm takes as input a security parameter λ , and returns as output the public parameters obtained from the configuration and setup of the infrastructure. Finally, the obtained public parameters (PP): the cloud server information ($CLParam$), the blockchain information and contracts deployed on it ($BLParam$), and the description of the bilinear group ($BMPParam$), are shared publicly for future interactions. More details on how to generate the outputs can be seen in Algorithm 1.

Algorithm 1 $\left(\overline{PP}^{\text{All}} \right) \leftarrow \text{Setup}(\lambda)$

- 1: Lunch a cloud storage provider to store data, and save its public parameters in $CLParam$
 - 2: Setup a blockchain platform with one smart contract, called Judge Contract (Jc), to audit interactions, and save its public parameters in $BLParam$
 - 3: $BMPParam = (\mathbb{G}_1, \mathbb{G}_T, q, g_1, g_T, e) \leftarrow \text{GroupGen}(\lambda)$
 - 4: $PP = (\lambda, BMPParam, BLParam, CLParam)$
 - 5: Output PP
-

Register phase. In this phase, there is only one algorithm named Register, which entities Fog, IoT device, and Client

can register in the system by calling it. It should be noted that this algorithm is defined uniformly for all of the aforementioned entities, however, it returns appropriate parameters for each entity according to the role parameter that it receives as input. More precisely, this algorithm is executed by TA, it receives as input the identity (id) and role of the entity and returns as output its private (sp^{id}) and public (pk^{id}) parameters. Finally, the public parameters of each registered entity are shared publicly. See Algorithm 2 for more details.

Algorithm 2 $\underbrace{(pk^{id}, sp^{id})}_{All} \leftarrow \text{Register}_{PP}(role, id)$

- 1: Parse PP as $(\lambda, BParam, BLParm, CLParam)$
 - 2: $k \leftarrow \text{SKE.Gen}(\lambda)$
 - 3: $(pk, sk) \leftarrow \text{PKE.Gen}(\lambda)$
 - 4: **if** ($role \in \{\text{Fog}, \text{Client}\}$) **then**
 - 5: $\text{SmartRegister}(pk, BLParm, role)$
 - 6: **Output** $(pk^{id}, sp^{id} = (k, sk))$
-

Data outsource phase. In this phase, three entities Fog, IoT device and Cloud interact with each other through Algorithms Collect and Outsource. In the following, a detailed description of each algorithm is given.

In the first step of the data outsourcing phase, the IoT devices execute Algorithm Collect to prepare the sensed data and send it to Fog. This algorithm takes as input the public key of the PKE encryption scheme (pk^F), the sensed data (m), the time window (wt), and the IoT identifier (id), and returns as output the encrypted data ($c_m^{wt,id}$) and the encrypted key of this encryption scheme ($c_k^{wt,id}$). Finally, both computed values are sent to Fog. It should be noted that to create a final file within a specific time window, the above procedure must be repeated several times in interaction with Fog. For more details, refer to Algorithm 3.

Algorithm 3 $\underbrace{(c_m^{wt,id}, c_k^{wt,id})}_{Fog} \leftarrow \text{Collect}_{sp}(pk^F, m, wt, id)$

- 1: Parse sp as (k, sk)
 - 2: $m' \leftarrow m \parallel H(m) \parallel \mathcal{DS}.\text{Sign}(sk, H(m)) \parallel ID \parallel wt$
 - 3: $c_m^{wt,id} \leftarrow \text{SKE.Enc}(k, m')$
 - 4: $c_k^{wt,id} \leftarrow \text{PKE.Enc}_{pk^F}(k)$
 - 5: **Output** $(c_m^{wt,id}, c_k^{wt,id})$
-

In the second step of the data outsourcing phase, Fog receives a set of encrypted data chunks for the same time window from the IoT device, and using Algorithm Outsource, aggregates them into an encrypted file and uploads it to the cloud. More precisely, Algorithm Outsource receives as input a bilinear group description ($BParam$), an encrypted data/key pair ($C_m^{wt,id}, C_k^{wt,id}$), and returns as output the following parameters:

- $C^{wt,id}$: It is an encrypted file obtained by aggregating data chunks for a specific time window, and it is uploaded to the cloud server.
- $H(C^{wt,id}), H(c')$: These values represent Fog's commitment to the file and encryption key, respectively,

and they are shared publicly.

- $(c', r^{wt,id})$: These values are used to reveal Fog's commitments in the data retrieval phase, and they are kept secret in this step.

More details on how to generate the outputs can be seen in Algorithm 4.

Algorithm 4 $\left(\underbrace{C_m^{wt,id}}_{Cloud}, \overbrace{H(C^{wt,id}), H(c'), c', r^{wt,id}}^{All} \right) \leftarrow \text{Outsource}_{(sp,pk)}(BParam, (C_m^{wt,id}, C_k^{wt,id}))$

- 1: Parse $BParam$ as $(\mathbb{G}_1, \mathbb{G}_T, q, g_1, g_T, e)$, sp as (k, sk) , and initialize $M = \emptyset$
 - 2: Parse $(C_m^{wt,id}, C_k^{wt,id})$ as $(c_{m,i}^{wt,id}, c_{k,i}^{wt,id})_{i \in [n]}$
 - 3: **for** (each $i \in [n]$) **do**
 - 4: $k_m := \text{PKE.Dec}_{sk}(c_{k,i}^{wt,id})$
 - 5: $m' := \text{SKE.Dec}_{k_m}(c_{m,i}^{wt,id})$
 - 6: Parse m' as $m' \leftarrow m \parallel h \parallel \sigma \parallel ID \parallel wt$
 - 7: **if** ($H(m) = h$ and $\mathcal{DS}.\text{Verify}_{pk}(m, \sigma) == 1$) **then**
 - 8: Append m to M
 - 9: **else**
 - 10: Abort
 - 11: $C^{wt,id} \leftarrow \text{SKE.Enc}_k(M)$
 - 12: $r^{wt,id} \leftarrow \mathbb{Z}_q^*$
 - 13: $c' = \langle c_1, c_2 \rangle = \langle g_1^{sk \cdot r}, e(g_1, g_1^{r^{wt,id}}) \cdot k \rangle$
 - 14: **Output** $(C^{wt,id}, H(C^{wt,id}), H(c'), c', r^{wt,id})$
-

Data retrieval and auditing phase. In this phase, four entities Fog, Client, Cloud and Smart contract interact with each other through Algorithms RetrievalData, PubVerify and ExtData. In the following, a detailed description of each algorithm is given.

In the first step of the data retrieval phase, the Fog executes Algorithm RetrievalData to provide metadata of the key related to the decryption of the requested file. More precisely, this algorithm receives as input a bilinear group description ($BParam$), the client's public key (pk^C), a request req which includes the IoT device identifier (id) and the desired time window (wt), and a secret pair $(r^{wt,id}, c')$ which contains information about the file decryption key. Upon receiving the mentioned inputs, this algorithm returns as output the reproduced metadata c'' for the file decryption key that is sent to the client, as well as the commitment $H(c'')$ that is publicly shared. See Algorithm 5 for more details.

Algorithm 5 $\left(\underbrace{c''}_{Client}, \overbrace{H(c'')}^{All} \right) \leftarrow \text{RetrievalData}_{sp}(BParam, pk^C, req, r^{wt,id}, c')$

- 1: Parse $BParam$ as $(\mathbb{G}_1, \mathbb{G}_T, q, g_1, g_T, e)$, req as (id, wt) , and c' as $\langle c_1, c_2 \rangle$
 - 2: $k_{FC} \leftarrow (pk^C)^{\frac{1}{sk}}$
 - 3: $t \leftarrow \mathbb{Z}_q^*$
 - 4: $c'' = \langle c'_1, c'_2, c_2 \rangle = \langle (k_{FC}^{r^{wt,id}})^{\frac{1}{t}}, c_1^t, c_2 \rangle$
 - 5: **Output** $(c'', H(c''))$
-

In the second step of the data retrieval phase, the client executes Algorithm PubVerify. If for any reason (hostile or non-hostile), the client claims that the output of this algorithm indicates that the data received from Fog is invalid, then by disclosing the value c'' to all entities (in particular, the smart contract), they can verify the validity of this claim through Algorithm PubVerify. More precisely, this algorithm takes as input the metadata of the key and its associated commitment, and returns as output the validity of the received data. Algorithm 6 shows the validation procedure in detail.

Algorithm 6 $b := \text{PubVerify}(\widehat{c'', H(c'')})$

```

1: if ( $H(\widehat{c''}) == H(c'')$ ) then
2:    $b = \text{True}$ 
3: else
4:    $b = \text{False}$ 
5: Output  $b$ 

```

In the final step of the data retrieval phase, the client executes Algorithm ExtData and obtains the plaintext of the requested file. More precisely, this algorithm takes as input the metadata of the key (c'') and the encrypted file (C), and returns as output the plaintext of the file M . For more details on file extraction, see Algorithm 7.

Algorithm 7 $M \leftarrow \text{ExtData}_{sp}(c'', C)$

```

1: Parse  $c''$  as  $\langle c'_1, c'_2, c_2 \rangle$ 
2:  $k = \frac{c_2}{e(c'_1, c'_2)^{\frac{1}{sk}}}$ 
3:  $M := \text{SKE.Dec}_k(C)$ 
4: Output  $M$ 

```

5 SECURITY ANALYSIS OF FAIRSHARE

In this section, we investigate the security analysis of Scheme FairShare presented in Section 4. The authors of this scheme claim in a theorem in paper [15] that it satisfies security properties: data privacy, data authenticity, fairness and accountability. For the reader's focus and to make the paper self-contained, the theorem is given below. It should be noted that no formal proof of this theorem is given in Paper [15], and the efforts are limited to providing some evidence.

Theorem 1 (Section VI, page 6, [13]). *If the hash function H is collusion-resistant, the encryption schemes SKE, PKE are CPA-secure and used blockchain is resilient to tampering, then an adversary controlling at most two of the three parties, cannot compromise authenticity, privacy, accountability and fairness, except with negligible probability.*

In the following, by giving an efficient adversary and describing its strategy, we show that the security property fairness does not hold for the mentioned scheme.

The Attack scenario model. Now, we describe the attack scenario model that violates the claim made in theorem 1. More precisely, we show that it is possible for Fog to behave maliciously, and as a result the fairness property claimed in

FairShare protocol does not hold. The configuration of the attack scenario model is as follows:

- All entities, except the Fog, execute all steps of the protocol honestly.
- The Fog also acts honestly in all steps related to the outsourcing phase,
- However, during the data retrieval phase, the Fog tries to cheat some clients. In particular, the Fog provides invalid information to the victim clients, despite receiving payment for it, and there is no way for the clients to prove this fraud.

In such a configuration, the Fog acts as follows to cheat the victim client:

- In the outsourcing phase (Algorithm Outsource, Line 13), the value of c' is computed as follows:

$$c' = \langle c_1, c_2 \rangle = \langle g_1^{sk \cdot r}, e(g_1, g_1^{r_{wt, id}}) \cdot k \rangle,$$

- In the data retrieval phase (Algorithm RetrievalData, Lines 4, 5), the Fog sequentially computes the values $c'' = \langle c'_1, c'_2, c_2 \rangle = \langle (k_{FC}^{wt, id})^{\frac{1}{t}}, c_1^t, c_2 \rangle$ (for the client) and $H(c'')$ (for the public). As it can be seen in Algorithm PubVerify, the criterion for verifying the correctness of the fog operation is to check the relation ($H(\widehat{c''}) == H(c'')$) on the aforementioned values, which can be valid for any values that maintain the above relation.

In such a situation, choosing any value of $\langle r_1, r_2, r_3 \rangle$ in the allowed range for c'' , and then calculating its hash value can be considered a successful forgery for the Fog.

6 CONCLUSIONS AND FUTURE WORKS

In this paper, we have studied the secure data sharing schemes for IoT platforms. In this regard, we first described the scenario model of these schemes and showed that they enable IoT devices to securely share their sensed data in industrial environments with clients through an appropriate infrastructure and intermediary entities. We then introduced concepts such as: data privacy, data authenticity, fairness and accountability as some of the most important security features raised in response to the challenges presented to these schemes. In the following, we focused specifically on the scheme proposed in paper [15], which claims to cover all of the mentioned security features. Finally, we examined the security analysis of the aforementioned scheme and showed that, despite the claim, this scheme does not cover fairness property and therefore cannot be used as a valid scheme for real-world applications.

For future contributions, it is important to propose an efficiency scheme that covers all the mentioned features and that is a suitable alternative to the scheme discussed in this paper. Also, adding other security features such as anonymity, which have been discussed in other researches, can expand the range of use of these schemes in real-world applications.

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